

AE-02

Self-Evolving Autonomous Research Agent

Problem statement Build a secure research and computer-use agent that can browse, retrieve evidence, execute code, operate an isolated workspace, and improve its strategies over time without silently rewriting its trusted control layer.

AT A GLANCE

Design dimension	Participant expectation
Primary user	Researchers, analysts, and teams running multi-step investigations.
Core inputs	Question, source policies, browser/tool capabilities, documents, datasets, and prior project memory.
Required outputs	Evidence-backed report, citations, reproducible artifacts, trace, and improvement log.
Hardest challenge	Making self-improvement measurable while preventing unsafe drift and prompt-injection takeover.
Proof of quality	Longitudinal success curve, citation metrics, injection resistance, and rollback tests.

Core objective

Create an agent that turns an open-ended question into a reproducible research package through live retrieval, document analysis, code execution, browser navigation, and controlled computer interaction. The defining feature is governed self-evolution: the agent may propose better prompts, tool-selection policies, retrieval strategies, or reusable skills, but each change must be evaluated and approved before it becomes trusted behavior.

Required capabilities and system blueprint

- Research planner producing a graph of subquestions, evidence requirements, source classes, confidence thresholds, and stop conditions.
- Hybrid live RAG with keyword and dense retrieval, time-aware ranking, source freshness, deduplication, contradiction tracking, and provenance for every passage.
- Browser and file controller that treats webpages, PDFs, comments, advertisements, and downloads as untrusted input and defends against prompt injection.
- Sandboxed notebook/terminal workspace with disposable sessions, snapshots, resource quotas, network allowlists, download quarantine, and audit logs.
- Evidence graph linking claims to sources, timestamps, locations, transformations, and uncertainty; outputs must distinguish observation from interpretation.
- Controlled self-improvement loop that logs strategy outcomes, critiques failures, proposes a versioned change, tests it on held-out tasks, and supports rollback.
- Human approval gates for sensitive actions such as external account changes, publication, communication, deletion, payments, or access to private data.
- Long-term project memory for source summaries, contradictions, research methods, user preferences, and unresolved questions with expiry rules.

Minimum viable demonstration

1. Give the agent a complex question requiring at least three source types, one structured dataset, and one code-based calculation.
2. Inject a malicious instruction into a webpage or downloaded document and show that the agent ignores it while preserving useful evidence.
3. Let the agent propose a retrieval or tool-selection improvement, run it against held-out tasks, and compare old versus new strategy.
4. Produce a cited report, a source/evidence map, a reproducible data artifact, and an exportable action trace.

Evaluation and benchmarks

- GAIA, WebArena/VisualWebArena, OSWorld, PaperBench-style replication, citation validation, long-context synthesis, and organizer-provided live-information tasks.
- Measure end-to-end success, citation precision/recall, unsupported-claim rate, source quality, browser success, code-execution success, prompt-injection resistance, cost, and human interventions.

- Add a longitudinal benchmark across multiple sessions to show whether the self-evolution loop improves held-out performance rather than merely changing behavior.

Hard-mode extensions

- Contradictory sources, multilingual material, malformed scans, dynamic websites, large document collections, stale evidence, and tasks requiring browser plus terminal plus desktop tools.
- A self-proposed skill that attempts to expand its permissions, alter a policy file, or reach an unapproved network must be blocked and reported.

Safety boundary System control, scraping, and self-evolution are permitted only inside a disposable, capability-scoped workspace. Do not bypass CAPTCHAs, access private accounts without approval, exfiltrate credentials, or let generated skills modify trusted control code without review and rollback.

Expected deliverables

1. Research planner, retrieval/indexing pipeline, browser/desktop adapters, and sandbox service.
2. Evidence/citation graph, memory store, self-improvement evaluator, policy engine, and trace viewer.
3. Longitudinal benchmark with before/after strategy results and held-out tasks.
4. Threat model, prompt-injection tests, data-handling notes, and rollback demonstration.
5. Cited final report plus machine-readable research package.

Why this stands out

Most research agents stop at retrieval and summarization. This challenge asks whether an agent can learn better ways to research while remaining inspectable, reversible, and safe — a much harder claim than adding another search box to a chatbot.